

AGA CoStays

Complete Backend Microservices Architecture and Package Structure

Regenerated master reference for one-by-one microservice ZIP generation

Document Scope

This document defines the backend architecture, workflow, application names, base packages, databases, responsibilities and complete package/file structures for all 14 AGA CoStays microservices. It is intended to be provided to a coding agent together with instructions to generate one complete microservice ZIP at a time.

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1. Application Overview

AGA CoStays is a multi-branch hotel and restaurant management platform supporting public browsing, customer bookings, room operations, restaurant orders, payments, billing, notifications, employee attendance, payroll, analytics and customer support.

2. End-to-End Backend Workflow

React Frontend
-> API Gateway Service
-> JWT authentication and role / permission / branch / ownership validation
-> Target business microservice
-> PostgreSQL database owned by that microservice
-> Synchronous OpenFeign/WebClient validation where an immediate answer is required
-> Apache Kafka events for notifications, billing updates, analytics and background workflows
-> Docker/Kubernetes deployment with Actuator, Prometheus, Grafana and centralized logs

3. Technology Stack

Java 21 LTS
Spring Boot 4.1.x and compatible Spring Cloud release train
Spring Web, Spring Validation, Spring Security, Spring Data JPA, Hibernate
Spring Cloud Gateway, OpenFeign and WebClient
PostgreSQL database per service, Redis where required, Flyway migrations
Apache Kafka 4.3.x in KRaft mode and Spring Kafka
Lombok, Swagger/OpenAPI, Maven
Docker, Docker Compose, Kubernetes
Actuator, Micrometer, Prometheus, Grafana, OpenSearch/ELK, OpenTelemetry
JUnit, Mockito, MockMvc, Testcontainers, Postman/Newman

4. Microservice Catalog

No.	Service	Application	Base Package	Database
1	API Gateway Service	api-gateway-service	com.agacostays.gateway	No database
2	Auth Service	auth-service	com.agacostays.auth	auth_db
3	User Service	user-service	com.agacostays.user	user_db
4	Branch Service	branch-service	com.agacostays.branch	branch_db
5	Room Service	room-service	com.agacostays.room	room_db
6	Booking Service	booking-service	com.agacostays.booking	booking_db
7	Restaurant Service	restaurant-service	com.agacostays.restaurant	restaurant_db
8	Payment Service	payment-service	com.agacostays.payment	payment_db
9	Billing Service	billing-service	com.agacostays.billing	billing_db
10	Notification Service	notification-service	com.agacostays.notification	notification_db
11	Attendance Service	attendance-service	com.agacostays.attendance	attendance_db
12	Payroll Service	payroll-service	com.agacostays.payroll	payroll_db
13	Analytics Service	analytics-service	com.agacostays.analytics	analytics_db
14	Support Feedback Service	support-feedback-service	com.agacostays.support	support_db

5. Shared Architecture Rules

1. Every business service owns its database and migration scripts.
2. Controllers use request/response DTOs and never expose entities directly.
3. Business logic remains in service implementations.
4. Enumerated fields use EnumType.STRING.
5. Security is validated at gateway and again inside each downstream service.
6. Immediate validation uses REST/OpenFeign/WebClient; async workflows use Kafka.
7. Every request and event carries traceId/correlationId.
8. ApiResponse and GlobalExceptionHandler maintain consistent responses.
9. Secrets are environment variables or Kubernetes Secrets, never hard-coded.
10. Each microservice contains Docker, configuration, migrations and mirrored tests.

6. Common Root Files

```
<microservice>/
pom.xml
mvnw
mvnw.cmd
.mvn/wrapper/maven-wrapper.properties
Dockerfile
.dockignore
README.md
src/main/java/<base-package>/...
src/main/resources/application.yml
src/main/resources/application-dev.yml
src/main/resources/application-prod.yml
src/main/resources/logback-spring.xml
src/main/resources/db/migration/...
src/test/java/<base-package>/...
```

1. API Gateway Service

Field	Value
Application folder	api-gateway-service
Application class	ApiGatewayApplication.java
Base package	com.agacostays.gateway
Database	No database
Purpose	Single entry point, routing, JWT filtering, CORS, rate limiting, trace IDs and gateway errors

Workflow

Frontend request -> Trace ID -> CORS/rate limit -> JWT validation for protected route -> claims forwarded -> target service -> response.

Package Structure

```
api-gateway-service/  
  pom.xml  
  mvnw  
  mvnw.cmd  
  .mvn/wrapper/maven-wrapper.properties  
  Dockerfile  
  .dockerignore  
  README.md  
  src/main/java/com/agacostays/gateway/  
    controller/  
      HealthController.java  
    service/  
      GatewayAuthService.java  
      GatewayRouteService.java  
    service/impl/  
      GatewayAuthServiceImpl.java  
      GatewayRouteServiceImpl.java  
    dto/response/  
      ApiResponse.java  
      GatewayErrorResponse.java  
      GatewayHealthResponse.java  
      RouteInfoResponse.java  
    exception/  
      GatewayExceptionHandler.java  
      GatewayUnauthorizedException.java  
      GatewayAccessDeniedException.java  
      GatewayRouteException.java  
      InvalidJwtTokenException.java  
      ExpiredJwtTokenException.java  
      MissingAuthorizationHeaderException.java  
      DownstreamServiceUnavailableException.java  
      RateLimitExceededException.java  
    security/  
      JwtTokenProvider.java  
      JwtClaims.java  
      PublicRouteValidator.java  
      RoleRouteValidator.java  
      GatewayAuthenticationEntryPoint.java  
    config/  
      GatewayRouteConfig.java  
      SecurityConfig.java  
      CorsConfig.java  
      RateLimitConfig.java  
      ActuatorConfig.java  
    filter/  
      JwtGatewayAuthenticationFilter.java  
      TraceIdFilter.java  
      RequestLoggingFilter.java  
      RateLimitFilter.java  
    constants/  
      GatewayConstants.java  
      RouteConstants.java  
      HeaderConstants.java  
      RoleConstants.java  
    util/  
      TraceIdUtil.java  
      JwtUtil.java  
      RequestUtil.java  
    repository/  
      .gitkeep  
    entity/  
      .gitkeep  
    dto/request/  
      .gitkeep  
    mapper/  
      .gitkeep  
    client/  
      .gitkeep  
    event/  
      .gitkeep  
  src/main/resources/  
    application.yml  
    application-dev.yml  
    application-prod.yml  
    logback-spring.xml  
    db/migration/  
      V1__create_core_tables.sql
```

```
V2__create_indexes.sql
V3__insert_default_data.sql
src/test/java/com/agacostays/gateway/
controller/
service/

repository/
mapper/
security/
producer/
consumer/
```

2. Auth Service

Field	Value
Application folder	auth-service
Application class	AuthServiceApplication.java
Base package	com.agacostays.auth
Database	auth_db
Purpose	Registration, login, JWT, refresh tokens, password reset, logout and OTP verification

Workflow

Register/login request -> validation -> user/role lookup -> password verification -> JWT/refresh token -> authentication response -> Kafka notification event.

Package Structure

```
auth-service/  
  pom.xml  
  mvnw  
  mvnw.cmd  
  .mvn/wrapper/maven-wrapper.properties  
  Dockerfile  
  .dockerignore  
  README.md  
  src/main/java/com/agacostays/auth/  
    controller/  
      AuthController.java  
      PasswordController.java  
      TokenController.java  
    service/  
      AuthService.java  
      JwtService.java  
      PasswordService.java  
      RefreshTokenService.java  
      OtpService.java  
    service/impl/  
      AuthServiceImpl.java  
      JwtServiceImpl.java  
      PasswordServiceImpl.java  
      RefreshTokenServiceImpl.java  
      OtpServiceImpl.java  
    repository/  
      UserRepository.java  
      RoleRepository.java  
      RefreshTokenRepository.java  
      PasswordResetTokenRepository.java  
      OtpTokenRepository.java  
    entity/  
      User.java  
      Role.java  
      RefreshToken.java  
      PasswordResetToken.java  
      OtpToken.java  
    dto/request/  
      RegisterRequest.java  
      LoginRequest.java  
      LogoutRequest.java  
      ForgotPasswordRequest.java  
      ResetPasswordRequest.java  
      ChangePasswordRequest.java  
      RefreshTokenRequest.java  
      ValidateTokenRequest.java  
      VerifyOtpRequest.java  
    dto/response/  
      ApiResponse.java  
      AuthResponse.java  
      TokenResponse.java  
      UserAuthResponse.java  
      PasswordResetResponse.java  
      OtpResponse.java  
    mapper/  
      UserAuthMapper.java  
      TokenMapper.java  
      OtpMapper.java  
    enums/  
      AuthProvider.java  
      TokenType.java  
      TokenStatus.java  
      OtpPurpose.java  
      LoginStatus.java  
      PasswordResetStatus.java  
    exception/  
      GlobalExceptionHandler.java  
      InvalidCredentialsException.java  
      UserAlreadyExistsException.java  
      UserNotFoundException.java  
      InvalidTokenException.java  
      TokenExpiredException.java  
      RefreshTokenException.java  
      PasswordResetException.java  
      OtpVerificationException.java  
    security/  
      AuthSecurityConfig.java
```

```

JwtTokenProvider.java
JwtAuthenticationFilter.java
CustomUserDetailsService.java
AuthenticationEntryPointHandler.java
AccessDeniedHandlerImpl.java

config/
  OpenApiConfig.java
  JpaAuditingConfig.java
  PasswordEncoderConfig.java
  KafkaProducerConfig.java
  KafkaTopicConfig.java
  AppConfig.java
client/
  UserServiceClient.java
  NotificationServiceClient.java
event/
  UserRegisteredEvent.java
  LoginSuccessEvent.java
  LoginFailedEvent.java
  PasswordResetRequestedEvent.java
producer/
  AuthEventProducer.java
consumer/
  .gitkeep
constants/
  AuthConstants.java
  JwtConstants.java
  KafkaTopicConstants.java
  HeaderConstants.java
util/
  JwtUtil.java
  PasswordUtil.java
  OtpUtil.java
  TokenUtil.java
src/main/resources/
  application.yml
  application-dev.yml
  application-prod.yml
  logback-spring.xml
db/migration/
  V1__create_core_tables.sql
  V2__create_indexes.sql
  V3__insert_default_data.sql
src/test/java/com/agacostays/auth/
  controller/
  service/
  repository/
  mapper/
  security/
  producer/
  consumer/

```

3. User Service

Field	Value
Application folder	user-service
Application class	UserServiceApplication.java
Base package	com.agacostays.user
Database	user_db
Purpose	Root Admin, Managers, Staff, Customers, Roles, Permissions and Staff-Branch mapping

Workflow

Admin/manager request -> role and branch validation -> user/profile/role persistence -> staff-branch mapping -> Kafka event.

Package Structure

```
user-service/  
  pom.xml  
  mvnw  
  mvnw.cmd  
  .mvn/wrapper/maven-wrapper.properties  
  Dockerfile  
  .dockerignore  
  README.md  
  src/main/java/com/agacostays/user/  
    controller/  
      RootAdminController.java  
      ManagerController.java  
      StaffController.java  
      CustomerController.java  
      RoleController.java  
      PermissionController.java  
      StaffBranchController.java  
      ProfileController.java  
    service/  
      RootAdminService.java  
      ManagerService.java  
      StaffService.java  
      CustomerService.java  
      RoleService.java  
      PermissionService.java  
      StaffBranchService.java  
      ProfileService.java  
    service/impl/  
      RootAdminServiceImpl.java  
      ManagerServiceImpl.java  
      StaffServiceImpl.java  
      CustomerServiceImpl.java  
      RoleServiceImpl.java  
      PermissionServiceImpl.java  
      StaffBranchServiceImpl.java  
      ProfileServiceImpl.java  
    repository/  
      UserRepository.java  
      RootAdminProfileRepository.java  
      ManagerProfileRepository.java  
      StaffRepository.java  
      CustomerRepository.java  
      RoleRepository.java  
      PermissionRepository.java  
      RolePermissionRepository.java  
      StaffBranchMappingRepository.java  
    entity/  
      User.java  
      RootAdminProfile.java  
      ManagerProfile.java  
      Staff.java  
      Customer.java  
      Role.java  
      Permission.java  
      RolePermission.java  
      StaffBranchMapping.java  
    dto/request/  
      CreateManagerRequest.java  
      UpdateManagerRequest.java  
      CreateStaffRequest.java  
      UpdateStaffRequest.java  
      AssignRoleRequest.java  
      CreateRoleRequest.java  
      UpdateRoleRequest.java  
      CreatePermissionRequest.java  
      AssignPermissionRequest.java  
      AssignStaffToBranchRequest.java  
      TransferBranchRequest.java  
      UpdateCustomerStatusRequest.java  
      UpdateUserStatusRequest.java  
      UpdateProfileRequest.java  
      ChangePasswordRequest.java  
    dto/response/  
      ApiResponse.java  
      UserResponse.java  
      UserProfileResponse.java  
      RootAdminProfileResponse.java  
      ManagerResponse.java  
      StaffResponse.java
```

```

CustomerResponse.java
RoleResponse.java
PermissionResponse.java

RolePermissionResponse.java
StaffBranchMappingResponse.java
StaffBranchResponse.java
UserStatusResponse.java
BranchResponse.java
mapper/
  UserMapper.java
  RootAdminMapper.java
  ManagerMapper.java
  StaffMapper.java
  CustomerMapper.java
  RoleMapper.java
  PermissionMapper.java
  RolePermissionMapper.java
  StaffBranchMappingMapper.java
enums/
  UserType.java
  UserStatus.java
  ManagerAccessLevel.java
  StaffStatus.java
  CustomerStatus.java
  RoleStatus.java
  Department.java
  Shift.java
  BranchAssignmentStatus.java
  StaffTransferStatus.java
  PermissionAction.java
  PermissionModule.java
exception/
  GlobalExceptionHandler.java
  UserNotFoundException.java
  RoleNotFoundException.java
  PermissionNotFoundException.java
  InvalidRoleException.java
  DuplicateUserException.java
  DuplicateRoleException.java
  DuplicatePermissionException.java
  BranchMappingException.java
  StaffNotFoundException.java
  CustomerNotFoundException.java
  ManagerNotFoundException.java
  RootAdminNotFoundException.java
  InvalidUserStatusException.java
  InvalidStaffTransferException.java
  PermissionDeniedException.java
  InvalidCredentialsException.java
security/
  UserSecurityConfig.java
  JwtAuthenticationFilter.java
  CurrentUserProvider.java
  PermissionValidator.java
  BranchAccessValidator.java
  RoleBasedAccessService.java
  MethodSecurityConfig.java
  AuthenticationEntryPointHandler.java
  AccessDeniedHandlerImpl.java
config/
  OpenApiConfig.java
  JpaAuditingConfig.java
  PasswordEncoderConfig.java
  FeignClientConfig.java
  KafkaProducerConfig.java
  KafkaTopicConfig.java
  AppConfig.java
client/
  BranchServiceClient.java
  NotificationServiceClient.java
event/
  ManagerCreatedEvent.java
  StaffCreatedEvent.java
  StaffAssignedToBranchEvent.java
  StaffTransferredEvent.java
  RoleAssignedEvent.java
  CustomerStatusChangedEvent.java
producer/
  UserRoleEventProducer.java
consumer/
  .gitkeep
constants/
  UserRoleConstants.java
  PermissionConstants.java
  RoleConstants.java

  UserRouteConstants.java
  KafkaTopicConstants.java
  HeaderConstants.java
util/
  UserUtil.java
  RoleUtil.java
  PermissionUtil.java
  PasswordUtil.java
  MaskingUtil.java
validation/
  UserValidationService.java
  RoleValidationService.java
  StaffValidationService.java
  CustomerValidationService.java
  BranchMappingValidationService.java
audit/
  AuditLogRequest.java
  UserAuditHelper.java

```

```
src/main/resources/  
  application.yml  
  application-dev.yml  
  application-prod.yml  
  logback-spring.xml  
  db/migration/  
    V1__create_core_tables.sql  
    V2__create_indexes.sql  
    V3__insert_default_data.sql  
src/test/java/com/agacostays/user/  
  controller/  
  service/  
  repository/  
  mapper/  
  security/  
  producer/  
  consumer/
```

4. Branch Service

Field	Value
Application folder	branch-service
Application class	BranchServiceApplication.java
Base package	com.agacostays.branch
Database	branch_db
Purpose	Cities, hotel branches, photos, receptionist contacts and emergency contacts

Workflow

City/branch request -> validation -> branch and photo persistence -> contact retrieval -> public branch response -> Kafka update event.

Package Structure

```
branch-service/
  pom.xml
  mvnw
  mvnw.cmd
  .mvn/wrapper/maven-wrapper.properties
  Dockerfile
  .dockerignore
  README.md
  src/main/java/com/agacostays/branch/
    controller/
      CityController.java
      HotelBranchController.java
      HotelBranchPhotoController.java
      ReceptionistContactController.java
      EmergencyContactController.java
    service/
      CityService.java
      HotelBranchService.java
      HotelBranchPhotoService.java
      ReceptionistContactService.java
      EmergencyContactService.java
    service/impl/
      CityServiceImpl.java
      HotelBranchServiceImpl.java
      HotelBranchPhotoServiceImpl.java
      ReceptionistContactServiceImpl.java
      EmergencyContactServiceImpl.java
    repository/
      CityRepository.java
      HotelBranchRepository.java
      HotelBranchPhotoRepository.java
      ReceptionistContactRepository.java
      EmergencyContactRepository.java
    entity/
      City.java
      HotelBranch.java
      HotelBranchPhoto.java
      ReceptionistContact.java
      EmergencyContact.java
    dto/request/
      CreateCityRequest.java
      UpdateCityRequest.java
      HotelBranchRequest.java
      UpdateHotelBranchRequest.java
      HotelBranchPhotoRequest.java
      ReceptionistContactRequest.java
      EmergencyContactRequest.java
      UpdateBranchStatusRequest.java
      UpdateContactStatusRequest.java
    dto/response/
      ApiResponse.java
      CityResponse.java
      HotelBranchResponse.java
      HotelBranchSummaryResponse.java
      HotelBranchPhotoResponse.java
      ReceptionistContactResponse.java
      EmergencyContactResponse.java
      BranchContactResponse.java
      BranchPublicResponse.java
      UserResponse.java
    mapper/
      CityMapper.java
      HotelBranchMapper.java
      HotelBranchPhotoMapper.java
      ReceptionistContactMapper.java
      EmergencyContactMapper.java
    enums/
      CityStatus.java
      BranchStatus.java
      BranchPhotoType.java
      ContactType.java
      ContactAvailabilityStatus.java
      EmergencyContactPurpose.java
    exception/
      GlobalExceptionHandler.java
      CityNotFoundException.java
      BranchNotFoundException.java
```

```

BranchPhotoNotFoundException.java
BranchPhotoException.java
ReceptionistContactNotFoundException.java
EmergencyContactNotFoundException.java
InvalidLocationException.java

DuplicateCityException.java
DuplicateBranchException.java
InvalidBranchStatusException.java
InvalidContactStatusException.java
FileUploadException.java
BranchAccessDeniedException.java
security/
BranchSecurityConfig.java
JwtAuthenticationFilter.java
CurrentUserProvider.java
BranchAccessValidator.java
FileUploadSecurityValidator.java
RoleBasedAccessService.java
MethodSecurityConfig.java
AuthenticationEntryPointHandler.java
AccessDeniedHandlerImpl.java
config/
OpenApiConfig.java
JpaAuditingConfig.java
FeignClientConfig.java
KafkaProducerConfig.java
KafkaTopicConfig.java
FileStorageConfig.java
AppConfig.java
client/
UserServiceClient.java
NotificationServiceClient.java
event/
BranchCreatedEvent.java
BranchUpdatedEvent.java
BranchStatusChangedEvent.java
BranchPhotoUpdatedEvent.java
ReceptionistContactUpdatedEvent.java
EmergencyContactUpdatedEvent.java
producer/
BranchEventProducer.java
consumer/
.gitkeep
constants/
BranchConstants.java
CityConstants.java
BranchRouteConstants.java
FileConstants.java
KafkaTopicConstants.java
HeaderConstants.java
util/
LocationUtil.java
FileUploadUtil.java
BranchUtil.java
ContactUtil.java
validation/
CityValidationService.java
BranchValidationService.java
LocationValidationService.java
BranchPhotoValidationService.java
ContactValidationService.java
audit/
AuditLogRequest.java
BranchAuditHelper.java
src/main/resources/
application.yml
application-dev.yml
application-prod.yml
logback-spring.xml
db/migration/
V1__create_core_tables.sql
V2__create_indexes.sql
V3__insert_default_data.sql
src/test/java/com/agacostays/branch/
controller/
service/
repository/
mapper/
security/
producer/
consumer/

```

5. Room Service

Field	Value
Application folder	room-service
Application class	RoomServiceApplication.java
Base package	com.agacostays.room
Database	room_db
Purpose	Rooms, photos, room state, availability and pricing history

Workflow

Room request -> branch validation -> availability/status/price handling -> persistence -> price history -> Kafka room event.

Package Structure

```
room-service/  
  pom.xml  
  mvnw  
  mvnw.cmd  
  .mvn/wrapper/maven-wrapper.properties  
  Dockerfile  
  .dockerignore  
  README.md  
  src/main/java/com/agacostays/room/  
    controller/  
      RoomController.java  
      RoomPhotoController.java  
      RoomPriceController.java  
      RoomAvailabilityController.java  
      RoomStatusController.java  
    service/  
      RoomService.java  
      RoomPhotoService.java  
      RoomPriceService.java  
      RoomAvailabilityService.java  
      RoomStatusService.java  
    service/impl/  
      RoomServiceImpl.java  
      RoomPhotoServiceImpl.java  
      RoomPriceServiceImpl.java  
      RoomAvailabilityServiceImpl.java  
      RoomStatusServiceImpl.java  
    repository/  
      RoomRepository.java  
      RoomPhotoRepository.java  
      RoomPriceHistoryRepository.java  
    entity/  
      Room.java  
      RoomPhoto.java  
      RoomPriceHistory.java  
    dto/request/  
      CreateRoomRequest.java  
      UpdateRoomRequest.java  
      UpdateRoomPriceRequest.java  
      UpdateRoomStatusRequest.java  
      RoomAvailabilityRequest.java  
      RoomPhotoRequest.java  
      DeleteRoomPhotoRequest.java  
      RoomSearchRequest.java  
    dto/response/  
      ApiResponse.java  
      RoomResponse.java  
      RoomSummaryResponse.java  
      RoomPhotoResponse.java  
      RoomPriceHistoryResponse.java  
      RoomAvailabilityResponse.java  
      RoomStatusResponse.java  
      BranchResponse.java  
      BookingStatusResponse.java  
    mapper/  
      RoomMapper.java  
      RoomPhotoMapper.java  
      RoomPriceHistoryMapper.java  
    enums/  
      RoomType.java  
      RoomStatus.java  
      RoomPhotoType.java  
      RoomAvailabilityStatus.java  
      RoomPriceChangeReason.java  
      RoomFacility.java  
    exception/  
      GlobalExceptionHandler.java  
      RoomNotFoundException.java  
      RoomNotAvailableException.java  
      RoomAlreadyExistsException.java  
      RoomPriceUpdateException.java  
      InvalidRoomStatusException.java  
      RoomPhotoNotFoundException.java  
      RoomPhotoException.java  
      BranchNotFoundException.java  
      BranchAccessDeniedException.java  
      FileUploadException.java  
      ActiveBookingFoundException.java  
    security/
```

```

RoomSecurityConfig.java
JwtAuthenticationFilter.java
CurrentUserProvider.java

BranchAccessValidator.java
FileUploadSecurityValidator.java
RoleBasedAccessService.java
MethodSecurityConfig.java
AuthenticationEntryPointHandler.java
AccessDeniedHandlerImpl.java
config/
  OpenApiConfig.java
  JpaAuditingConfig.java
  FeignClientConfig.java
  KafkaProducerConfig.java
  KafkaTopicConfig.java
  FileStorageConfig.java
  AppConfig.java
client/
  BranchServiceClient.java
  BookingServiceClient.java
event/
  RoomCreatedEvent.java
  RoomUpdatedEvent.java
  RoomDeletedEvent.java
  RoomPriceUpdatedEvent.java
  RoomStatusChangedEvent.java
  RoomPhotoUploadedEvent.java
  RoomPhotoDeletedEvent.java
producer/
  RoomEventProducer.java
consumer/
  .gitkeep
constants/
  RoomConstants.java
  RoomRouteConstants.java
  RoomStatusConstants.java
  FileConstants.java
  KafkaTopicConstants.java
  HeaderConstants.java
util/
  RoomUtil.java
  RoomPriceUtil.java
  RoomAvailabilityUtil.java
  FileUploadUtil.java
  DateRangeUtil.java
validation/
  RoomValidationService.java
  RoomPhotoValidationService.java
  RoomPriceValidationService.java
  RoomStatusValidationService.java
  RoomAvailabilityValidationService.java
audit/
  AuditLogRequest.java
  RoomAuditHelper.java
src/main/resources/
  application.yml
  application-dev.yml
  application-prod.yml
  logback-spring.xml
db/migration/
  V1__create_core_tables.sql
  V2__create_indexes.sql
  V3__insert_default_data.sql
src/test/java/com/agacostays/room/
  controller/
  service/
  repository/
  mapper/
  security/
  producer/
  consumer/

```

6. Booking Service

Field	Value
Application folder	booking-service
Application class	BookingServiceApplication.java
Base package	com.agacostays.booking
Database	booking_db
Purpose	Bookings, approvals, cancellations, check-in, checkout and booking events

Workflow

Booking request -> customer/room validation -> availability and price snapshot -> booking persistence -> approval/check-in/checkout -> Kafka event.

Package Structure

```
booking-service/  
  pom.xml  
  mvnw  
  mvnw.cmd  
  .mvn/wrapper/maven-wrapper.properties  
  Dockerfile  
  .dockerignore  
  README.md  
  src/main/java/com/agacostays/booking/  
    controller/  
      BookingController.java  
      CustomerBookingController.java  
      CheckInController.java  
      CheckOutController.java  
      BookingStatusController.java  
    service/  
      BookingService.java  
      BookingValidationService.java  
      BookingPriceService.java  
      CheckInService.java  
      CheckOutService.java  
      BookingNotificationService.java  
    service/impl/  
      BookingServiceImpl.java  
      BookingValidationServiceImpl.java  
      BookingPriceServiceImpl.java  
      CheckInServiceImpl.java  
      CheckOutServiceImpl.java  
      BookingNotificationServiceImpl.java  
    repository/  
      BookingRepository.java  
      BookingHistoryRepository.java  
    entity/  
      Booking.java  
      BookingHistory.java  
    dto/request/  
      CreateBookingRequest.java  
      UpdateBookingRequest.java  
      BookingStatusRequest.java  
      BookingCancelRequest.java  
      CheckInRequest.java  
      CheckOutRequest.java  
      BookingSearchRequest.java  
      BookingDateRangeRequest.java  
      CustomerBookingFilterRequest.java  
    dto/response/  
      ApiResponse.java  
      BookingResponse.java  
      BookingSummaryResponse.java  
      BookingHistoryResponse.java  
      CheckInResponse.java  
      CheckOutResponse.java  
      BookingStatusResponse.java  
      BookingPriceResponse.java  
      RoomResponse.java  
      CustomerResponse.java  
      BillingResponse.java  
      PaymentStatusResponse.java  
    mapper/  
      BookingMapper.java  
      BookingHistoryMapper.java  
    enums/  
      BookingStatus.java  
      BookingSource.java  
      BookingAction.java  
      CheckInStatus.java  
      CheckOutStatus.java  
      GuestIdProofType.java  
      CancellationReason.java  
      PaymentStatus.java  
    exception/  
      GlobalExceptionHandler.java  
      BookingNotFoundException.java  
      RoomNotAvailableException.java  
      BookingNotAllowedException.java  
      InvalidBookingStatusException.java  
      CustomerMismatchException.java
```

```

CustomerNotFoundException.java
RoomNotFoundException.java
BranchNotFoundException.java
BranchAccessDeniedException.java
CheckInNotAllowedException.java

CheckOutNotAllowedException.java
DuplicateBookingException.java
InvalidDateRangeException.java
PaymentRequiredException.java
security/
BookingSecurityConfig.java
JwtAuthenticationFilter.java
CurrentUserProvider.java
OwnerAccessValidator.java
BranchAccessValidator.java
RoleBasedAccessService.java
MethodSecurityConfig.java
AuthenticationEntryPointHandler.java
AccessDeniedHandlerImpl.java
config/
OpenApiConfig.java
JpaAuditingConfig.java
FeignClientConfig.java
KafkaProducerConfig.java
KafkaConsumerConfig.java
KafkaTopicConfig.java
AppConfig.java
client/
RoomServiceClient.java
UserServiceClient.java
BillingServiceClient.java
PaymentServiceClient.java
NotificationServiceClient.java
event/
BookingCreatedEvent.java
BookingApprovedEvent.java
BookingRejectedEvent.java
BookingCancelledEvent.java
CheckInCompletedEvent.java
CheckoutCompletedEvent.java
BookingPaymentStatusUpdatedEvent.java
BookingStatusChangedEvent.java
producer/
BookingEventProducer.java
consumer/
PaymentSuccessEventConsumer.java
PaymentFailedEventConsumer.java
RefundProcessedEventConsumer.java
BillingGeneratedEventConsumer.java
constants/
BookingConstants.java
BookingRouteConstants.java
BookingStatusConstants.java
KafkaTopicConstants.java
HeaderConstants.java
util/
BookingUtil.java
BookingPriceUtil.java
DateRangeUtil.java
BookingStatusUtil.java
GuestIdProofUtil.java
validation/
BookingDateValidationService.java
BookingStatusValidationService.java
BookingOwnershipValidationService.java
CheckInValidationService.java
CheckOutValidationService.java
RoomAvailabilityValidationService.java
audit/
AuditLogRequest.java
BookingAuditHelper.java
src/main/resources/
application.yml
application-dev.yml
application-prod.yml
logback-spring.xml
db/migration/
V1__create_core_tables.sql
V2__create_indexes.sql
V3__insert_default_data.sql
src/test/java/com/agacostays/booking/
controller/
service/
repository/
mapper/
security/
producer/

consumer/

```

7. Restaurant Service

Field	Value
Application folder	restaurant-service
Application class	RestaurantServiceApplication.java
Base package	com.agacostays.restaurant
Database	restaurant_db
Purpose	Restaurant, menu, dish photos, orders, kitchen and serving workflow

Workflow

Food order -> booking and branch validation -> order persistence -> accept -> prepare -> ready -> deliver -> billing and Kafka events.

Package Structure

```
restaurant-service/  
  pom.xml  
  mvnw  
  mvnw.cmd  
  .mvn/wrapper/maven-wrapper.properties  
  Dockerfile  
  .dockerignore  
  README.md  
  src/main/java/com/agacostays/restaurant/  
    controller/  
      RestaurantController.java  
      MenuController.java  
      DishPhotoController.java  
      FoodOrderController.java  
      KitchenController.java  
      ServingController.java  
      RestaurantFeedbackController.java  
      RestaurantAssignmentController.java  
    service/  
      RestaurantService.java  
      MenuService.java  
      DishPhotoService.java  
      FoodOrderService.java  
      KitchenService.java  
      ServingService.java  
      RestaurantFeedbackService.java  
      RestaurantOrderAssignmentService.java  
    service/impl/  
      RestaurantServiceImpl.java  
      MenuServiceImpl.java  
      DishPhotoServiceImpl.java  
      FoodOrderServiceImpl.java  
      KitchenServiceImpl.java  
      ServingServiceImpl.java  
      RestaurantFeedbackServiceImpl.java  
      RestaurantOrderAssignmentServiceImpl.java  
    repository/  
      RestaurantRepository.java  
      RestaurantMenuRepository.java  
      RestaurantMenuPhotoRepository.java  
      RestaurantOrderRepository.java  
      RestaurantOrderItemRepository.java  
      RestaurantFeedbackRepository.java  
      RestaurantOrderAssignmentRepository.java  
    entity/  
      Restaurant.java  
      RestaurantMenu.java  
      RestaurantMenuPhoto.java  
      RestaurantOrder.java  
      RestaurantOrderItem.java  
      RestaurantFeedback.java  
      RestaurantOrderAssignment.java  
    dto/request/  
      RestaurantRequest.java  
      CreateMenuItemRequest.java  
      UpdateMenuItemRequest.java  
      DishPhotoRequest.java  
      CreateFoodOrderRequest.java  
      UpdateFoodOrderStatusRequest.java  
      AssignChefRequest.java  
      AssignServingStaffRequest.java  
      RestaurantFeedbackRequest.java  
      MenuSearchRequest.java  
    dto/response/  
      ApiResponse.java  
      RestaurantResponse.java  
      MenuItemResponse.java  
      DishPhotoResponse.java  
      FoodOrderResponse.java  
      FoodOrderItemResponse.java  
      KitchenOrderResponse.java  
      ServingOrderResponse.java  
      RestaurantFeedbackResponse.java  
      BookingResponse.java  
      UserResponse.java  
      PaymentStatusResponse.java  
      BillingResponse.java
```

```

mapper/
  RestaurantMapper.java
  RestaurantMenuMapper.java
  RestaurantMenuPhotoMapper.java
  RestaurantOrderMapper.java

  RestaurantOrderItemMapper.java
  RestaurantFeedbackMapper.java
  RestaurantOrderAssignmentMapper.java
enums/
  RestaurantStatus.java
  MenuCategory.java
  FoodType.java
  MenuItemStatus.java
  FoodOrderStatus.java
  DeliveryType.java
  KitchenOrderStatus.java
  ServingOrderStatus.java
  RestaurantPaymentOption.java
  DishPhotoType.java
exception/
  GlobalExceptionHandler.java
  ResourceNotFoundException.java
  InvalidStatusException.java
  BusinessException.java
  AccessDeniedException.java
  DuplicateResourceException.java
security/
  RestaurantSecurityConfig.java
  JwtAuthenticationFilter.java
  CurrentUserProvider.java
  BranchAccessValidator.java
  RoleBasedAccessService.java
  MethodSecurityConfig.java
  AuthenticationEntryPointHandler.java
  AccessDeniedHandlerImpl.java
config/
  OpenApiConfig.java
  JpaAuditingConfig.java
  FeignClientConfig.java
  KafkaProducerConfig.java
  KafkaConsumerConfig.java
  KafkaTopicConfig.java
  AppConfig.java
client/
  BookingServiceClient.java
  UserServiceClient.java
  PaymentServiceClient.java
  BillingServiceClient.java
  NotificationServiceClient.java
event/
  FoodOrderPlacedEvent.java
  FoodOrderAcceptedEvent.java
  FoodOrderRejectedEvent.java
  FoodReadyEvent.java
  FoodDeliveredEvent.java
  FoodOrderCancelledEvent.java
producer/
  RestaurantEventProducer.java
consumer/
  PaymentSuccessEventConsumer.java
  PaymentFailedEventConsumer.java
  BookingCancelledEventConsumer.java
constants/
  RestaurantConstants.java
  KafkaTopicConstants.java
  HeaderConstants.java
util/
  RestaurantUtil.java
  DateRangeUtil.java
validation/
  RestaurantValidationService.java
audit/
  AuditLogRequest.java
  RestaurantAuditHelper.java
src/main/resources/
  application.yml
  application-dev.yml
  application-prod.yml
  logback-spring.xml
  db/migration/
    V1__create_core_tables.sql
    V2__create_indexes.sql
    V3__insert_default_data.sql
src/test/java/com/agacostays/restaurant/
  controller/
  service/
  repository/

  mapper/
  security/
  producer/
  consumer/

```

8. Payment Service

Field	Value
Application folder	payment-service
Application class	PaymentServiceApplication.java
Base package	com.agacostays.payment
Database	payment_db
Purpose	Payments, verification, gateway webhooks and refunds

Workflow

Payment request -> gateway order -> customer payment -> signature/webhook verification -> transaction persistence -> success/failure/refund event.

Package Structure

```
payment-service/  
  pom.xml  
  mvnw  
  mvnw.cmd  
  .mvn/wrapper/maven-wrapper.properties  
  Dockerfile  
  .dockerignore  
  README.md  
  src/main/java/com/agacostays/payment/  
    controller/  
      PaymentController.java  
      PaymentWebhookController.java  
      RefundController.java  
      RestaurantPaymentController.java  
    service/  
      PaymentService.java  
      PaymentGatewayService.java  
      PaymentVerificationService.java  
      RefundService.java  
      PaymentWebhookService.java  
    service/impl/  
      PaymentServiceImpl.java  
      PaymentGatewayServiceImpl.java  
      PaymentVerificationServiceImpl.java  
      RefundServiceImpl.java  
      PaymentWebhookServiceImpl.java  
    repository/  
      PaymentRepository.java  
      RefundRepository.java  
      PaymentTransactionRepository.java  
      PaymentWebhookLogRepository.java  
    entity/  
      Payment.java  
      Refund.java  
      PaymentTransaction.java  
      PaymentWebhookLog.java  
    dto/request/  
      CreatePaymentRequest.java  
      VerifyPaymentRequest.java  
      RefundRequest.java  
      PaymentWebhookRequest.java  
      RestaurantPaymentRequest.java  
      FinalBillPaymentRequest.java  
    dto/response/  
      ApiResponse.java  
      PaymentResponse.java  
      PaymentStatusResponse.java  
      VerifyPaymentResponse.java  
      RefundResponse.java  
      PaymentTransactionResponse.java  
      GatewayOrderResponse.java  
      BookingResponse.java  
      BillingResponse.java  
      RestaurantOrderResponse.java  
    mapper/  
      PaymentMapper.java  
      RefundMapper.java  
      PaymentTransactionMapper.java  
      PaymentWebhookLogMapper.java  
    enums/  
      PaymentStatus.java  
      PaymentFor.java  
      PaymentMethod.java  
      PaymentGatewayName.java  
      RefundStatus.java  
      RefundReason.java  
      WebhookEventType.java  
    exception/  
      GlobalExceptionHandler.java  
      ResourceNotFoundException.java  
      InvalidStatusException.java  
      BusinessException.java  
      AccessDeniedException.java  
      DuplicateResourceException.java  
    security/  
      PaymentSecurityConfig.java  
      JwtAuthenticationFilter.java
```

```

    CurrentUserProvider.java
    BranchAccessValidator.java
    RoleBasedAccessService.java
    MethodSecurityConfig.java
    AuthenticationEntryPointHandler.java

    AccessDeniedHandlerImpl.java
config/
    OpenApiConfig.java
    JpaAuditingConfig.java
    FeignClientConfig.java
    KafkaProducerConfig.java
    KafkaConsumerConfig.java
    KafkaTopicConfig.java
    AppConfig.java
client/
    BookingServiceClient.java
    BillingServiceClient.java
    RestaurantServiceClient.java
    NotificationServiceClient.java
event/
    PaymentSuccessEvent.java
    PaymentFailedEvent.java
    RefundProcessedEvent.java
    RefundFailedEvent.java
    PaymentWebhookReceivedEvent.java
producer/
    PaymentEventProducer.java
consumer/
    BookingCancelledEventConsumer.java
    FinalBillGeneratedEventConsumer.java
constants/
    PaymentConstants.java
    KafkaTopicConstants.java
    HeaderConstants.java
util/
    PaymentUtil.java
    DateRangeUtil.java
validation/
    PaymentValidationService.java
audit/
    AuditLogRequest.java
    PaymentAuditHelper.java
src/main/resources/
    application.yml
    application-dev.yml
    application-prod.yml
    logback-spring.xml
    db/migration/
        V1__create_core_tables.sql
        V2__create_indexes.sql
        V3__insert_default_data.sql
src/test/java/com/agacostays/payment/
    controller/
    service/
    repository/
    mapper/
    security/
    producer/
    consumer/

```

9. Billing Service

Field	Value
Application folder	billing-service
Application class	BillingServiceApplication.java
Base package	com.agacostays.billing
Database	billing_db
Purpose	Final bills, invoices, room charges and restaurant charges

Workflow

Checkout/order event -> collect room and restaurant charges -> tax/discount calculation -> bill persistence -> invoice generation -> payment status.

Package Structure

```
billing-service/  
  pom.xml  
  mvnw  
  mvnw.cmd  
  .mvn/wrapper/maven-wrapper.properties  
  Dockerfile  
  .dockerignore  
  README.md  
  src/main/java/com/agacostays/billing/  
    controller/  
      BillingController.java  
      FinalBillController.java  
      InvoiceController.java  
    service/  
      BillingService.java  
      FinalBillService.java  
      BillCalculationService.java  
      InvoiceService.java  
    service/impl/  
      BillingServiceImpl.java  
      FinalBillServiceImpl.java  
      BillCalculationServiceImpl.java  
      InvoiceServiceImpl.java  
    repository/  
      BillingRepository.java  
      InvoiceRepository.java  
      BillChargeRepository.java  
    entity/  
      Bill.java  
      Invoice.java  
      BillCharge.java  
    dto/request/  
      GenerateFinalBillRequest.java  
      PayFinalBillRequest.java  
      InvoiceGenerateRequest.java  
      BillSearchRequest.java  
      ApplyDiscountRequest.java  
    dto/response/  
      ApiResponse.java  
      FinalBillResponse.java  
      InvoiceResponse.java  
      BillCalculationResponse.java  
      BillChargeResponse.java  
      BookingResponse.java  
      RestaurantChargeResponse.java  
      PaymentStatusResponse.java  
    mapper/  
      BillMapper.java  
      InvoiceMapper.java  
      BillChargeMapper.java  
    enums/  
      BillStatus.java  
      BillType.java  
      InvoiceStatus.java  
      TaxType.java  
      DiscountType.java  
      ChargeType.java  
    exception/  
      GlobalExceptionHandler.java  
      ResourceNotFoundException.java  
      InvalidStatusException.java  
      BusinessException.java  
      AccessDeniedException.java  
      DuplicateResourceException.java  
    security/  
      BillingSecurityConfig.java  
      JwtAuthenticationFilter.java  
      CurrentUserProvider.java  
      BranchAccessValidator.java  
      RoleBasedAccessService.java  
      MethodSecurityConfig.java  
      AuthenticationEntryPointHandler.java  
      AccessDeniedHandlerImpl.java  
    config/  
      OpenApiConfig.java  
      JpaAuditingConfig.java  
      FeignClientConfig.java
```

```

KafkaProducerConfig.java
KafkaConsumerConfig.java
KafkaTopicConfig.java
AppConfig.java
client/

BookingServiceClient.java
RestaurantServiceClient.java
PaymentServiceClient.java
NotificationServiceClient.java
event/
FinalBillGeneratedEvent.java
InvoiceGeneratedEvent.java
BillPaidEvent.java
BillCancelledEvent.java
producer/
BillingEventProducer.java
consumer/
CheckoutCompletedEventConsumer.java
PaymentSuccessEventConsumer.java
FoodDeliveredEventConsumer.java
constants/
BillingConstants.java
KafkaTopicConstants.java
HeaderConstants.java
util/
BillingUtil.java
DateRangeUtil.java
validation/
BillingValidationService.java
audit/
AuditLogRequest.java
BillingAuditHelper.java
src/main/resources/
application.yml
application-dev.yml
application-prod.yml
logback-spring.xml
db/migration/
V1__create_core_tables.sql
V2__create_indexes.sql
V3__insert_default_data.sql
src/test/java/com/agacostays/billing/
controller/
service/
repository/
mapper/
security/
producer/
consumer/

```

10. Notification Service

Field	Value
Application folder	notification-service
Application class	NotificationServiceApplication.java
Base package	com.agacostays.notification
Database	notification_db
Purpose	Email, SMS, reminders, templates and logs

Workflow

Kafka/domain request -> template resolution -> email/SMS dispatch -> retry/error handling -> notification log persistence.

Package Structure

```
notification-service/  
  pom.xml  
  mvnw  
  mvnw.cmd  
  .mvn/wrapper/maven-wrapper.properties  
  Dockerfile  
  .dockerignore  
  README.md  
  src/main/java/com/agacostays/notification/  
    controller/  
      NotificationController.java  
      EmailController.java  
      SmsController.java  
      CheckoutReminderController.java  
      TemplateController.java  
    service/  
      NotificationService.java  
      EmailService.java  
      SmsService.java  
      TemplateService.java  
      CheckoutReminderService.java  
      NotificationLogService.java  
    service/impl/  
      NotificationServiceImpl.java  
      EmailServiceImpl.java  
      SmsServiceImpl.java  
      TemplateServiceImpl.java  
      CheckoutReminderServiceImpl.java  
      NotificationLogServiceImpl.java  
    repository/  
      NotificationLogRepository.java  
      CheckoutReminderRepository.java  
      EmailTemplateRepository.java  
      SmsTemplateRepository.java  
    entity/  
      NotificationLog.java  
      CheckoutReminder.java  
      EmailTemplate.java  
      SmsTemplate.java  
    dto/request/  
      SendEmailRequest.java  
      SendSmsRequest.java  
      BookingEmailRequest.java  
      CheckoutReminderRequest.java  
      FeedbackEmailRequest.java  
      TemplateRequest.java  
    dto/response/  
      ApiResponse.java  
      NotificationResponse.java  
      EmailResponse.java  
      SmsResponse.java  
      CheckoutReminderResponse.java  
      TemplateResponse.java  
    mapper/  
      NotificationLogMapper.java  
      CheckoutReminderMapper.java  
      EmailTemplateMapper.java  
      SmsTemplateMapper.java  
    enums/  
      NotificationType.java  
      NotificationChannel.java  
      NotificationStatus.java  
      EmailTemplateCode.java  
      SmsTemplateCode.java  
      ReminderStatus.java  
      ReadStatus.java  
    exception/  
      GlobalExceptionHandler.java  
      ResourceNotFoundException.java  
      InvalidStatusException.java  
      BusinessException.java  
      AccessDeniedException.java  
      DuplicateResourceException.java  
    security/  
      NotificationSecurityConfig.java  
      JwtAuthenticationFilter.java  
      CurrentUserProvider.java  
      BranchAccessValidator.java  
      RoleBasedAccessService.java
```

```

MethodSecurityConfig.java
AuthenticationEntryPointHandler.java
AccessDeniedHandlerImpl.java

config/
  OpenApiConfig.java
  JpaAuditingConfig.java
  FeignClientConfig.java
  KafkaProducerConfig.java
  KafkaConsumerConfig.java
  KafkaTopicConfig.java
  AppConfig.java
client/
  BookingServiceClient.java
  BranchServiceClient.java
  UserServiceClient.java
event/
  NotificationSentEvent.java
  NotificationFailedEvent.java
producer/
  NotificationEventProducer.java
consumer/
  BookingCreatedEventConsumer.java
  CheckoutCompletedEventConsumer.java
  PaymentSuccessEventConsumer.java
  SalaryCreditedEventConsumer.java
  FoodDeliveredEventConsumer.java
  SupportRequestCreatedEventConsumer.java
constants/
  NotificationConstants.java
  KafkaTopicConstants.java
  HeaderConstants.java
util/
  NotificationUtil.java
  DateRangeUtil.java
validation/
  NotificationValidationService.java
audit/
  AuditLogRequest.java
  NotificationAuditHelper.java
src/main/resources/
  application.yml
  application-dev.yml
  application-prod.yml
  logback-spring.xml
db/migration/
  V1__create_core_tables.sql
  V2__create_indexes.sql
  V3__insert_default_data.sql
src/test/java/com/agacostays/notification/
  controller/
  service/
  repository/
  mapper/
  security/
  producer/
  consumer/

```

11. Attendance Service

Field	Value
Application folder	attendance-service
Application class	AttendanceServiceApplication.java
Base package	com.agacostays.attendance
Database	attendance_db
Purpose	Staff attendance, check-in/out, corrections and reports

Workflow

Staff check-in/out -> user and branch validation -> duplicate validation -> attendance persistence -> monthly projection -> Kafka event.

Package Structure

```
attendance-service/  
  pom.xml  
  mvnw  
  mvnw.cmd  
  .mvn/wrapper/maven-wrapper.properties  
  Dockerfile  
  .dockerignore  
  README.md  
  src/main/java/com/agacostays/attendance/  
    controller/  
      AttendanceController.java  
      StaffAttendanceController.java  
      BranchAttendanceController.java  
      AttendanceReportController.java  
    service/  
      AttendanceService.java  
      AttendanceValidationService.java  
      WorkingStaffService.java  
      AttendanceReportService.java  
    service/impl/  
      AttendanceServiceImpl.java  
      AttendanceValidationServiceImpl.java  
      WorkingStaffServiceImpl.java  
      AttendanceReportServiceImpl.java  
    repository/  
      AttendanceRepository.java  
      AttendanceCorrectionRepository.java  
    entity/  
      Attendance.java  
      AttendanceCorrection.java  
    dto/request/  
      CheckInRequest.java  
      CheckOutRequest.java  
      MarkAttendanceRequest.java  
      AttendanceCorrectionRequest.java  
      AttendanceSearchRequest.java  
      MonthlyAttendanceRequest.java  
    dto/response/  
      ApiResponse.java  
      AttendanceResponse.java  
      CheckInResponse.java  
      CheckOutResponse.java  
      MonthlyAttendanceResponse.java  
      WorkingStaffResponse.java  
      AbsentStaffResponse.java  
      UserResponse.java  
    mapper/  
      AttendanceMapper.java  
      AttendanceCorrectionMapper.java  
    enums/  
      AttendanceStatus.java  
      AttendanceSource.java  
      CheckInOutStatus.java  
      LeaveType.java  
      AttendanceCorrectionStatus.java  
    exception/  
      GlobalExceptionHandler.java  
      ResourceNotFoundException.java  
      InvalidStatusException.java  
      BusinessException.java  
      AccessDeniedException.java  
      DuplicateResourceException.java  
    security/  
      AttendanceSecurityConfig.java  
      JwtAuthenticationFilter.java  
      CurrentUserProvider.java  
      BranchAccessValidator.java  
      RoleBasedAccessService.java  
      MethodSecurityConfig.java  
      AuthenticationEntryPointHandler.java  
      AccessDeniedHandlerImpl.java  
    config/  
      OpenApiConfig.java  
      JpaAuditingConfig.java  
      FeignClientConfig.java  
      KafkaProducerConfig.java  
      KafkaConsumerConfig.java
```

```

    KafkaTopicConfig.java
    AppConfig.java
client/
    UserServiceClient.java
    PayrollServiceClient.java

event/
    AttendanceMarkedEvent.java
    StaffCheckedInEvent.java
    StaffCheckedOutEvent.java
    AttendanceCorrectedEvent.java
producer/
    AttendanceEventProducer.java
consumer/
    .gitkeep
constants/
    AttendanceConstants.java
    KafkaTopicConstants.java
    HeaderConstants.java
util/
    AttendanceUtil.java
    DateRangeUtil.java
validation/
    AttendanceValidationService.java
audit/
    AuditLogRequest.java
    AttendanceAuditHelper.java
src/main/resources/
    application.yml
    application-dev.yml
    application-prod.yml
    logback-spring.xml
db/migration/
    V1__create_core_tables.sql
    V2__create_indexes.sql
    V3__insert_default_data.sql
src/test/java/com/agacostays/attendance/
    controller/
    service/
    repository/
    mapper/
    security/
    producer/
    consumer/

```

12. Payroll Service

Field	Value
Application folder	payroll-service
Application class	PayrollServiceApplication.java
Base package	com.agacostays.payroll
Database	payroll_db
Purpose	Salary structures, bank accounts, payroll, credits and salary slips

Workflow

Month payroll request -> active staff -> attendance summary -> salary structure -> gross/deductions/net -> payroll batch -> salary credit/slip event.

Package Structure

```
payroll-service/  
  pom.xml  
  mvnw  
  mvnw.cmd  
  .mvn/wrapper/maven-wrapper.properties  
  Dockerfile  
  .dockerignore  
  README.md  
  src/main/java/com/agacostays/payroll/  
    controller/  
      SalaryStructureController.java  
      StaffBankAccountController.java  
      PayrollController.java  
      SalaryCreditController.java  
      SalarySlipController.java  
      PayrollReportController.java  
      BranchPayrollController.java  
    service/  
      SalaryStructureService.java  
      StaffBankAccountService.java  
      PayrollService.java  
      PayrollCalculationService.java  
      SalaryCreditService.java  
      SalarySlipService.java  
      PayrollReportService.java  
    service/impl/  
      SalaryStructureServiceImpl.java  
      StaffBankAccountServiceImpl.java  
      PayrollServiceImpl.java  
      PayrollCalculationServiceImpl.java  
      SalaryCreditServiceImpl.java  
      SalarySlipServiceImpl.java  
      PayrollReportServiceImpl.java  
    repository/  
      SalaryStructureRepository.java  
      StaffBankAccountRepository.java  
      PayrollRepository.java  
      PayrollPaymentBatchRepository.java  
      SalarySlipRepository.java  
    entity/  
      SalaryStructure.java  
      StaffBankAccount.java  
      Payroll.java  
      PayrollPaymentBatch.java  
      SalarySlip.java  
    dto/request/  
      SalaryStructureRequest.java  
      StaffBankAccountRequest.java  
      GeneratePayrollRequest.java  
      CreditSalaryRequest.java  
      PayrollSearchRequest.java  
      SalarySlipRequest.java  
    dto/response/  
      ApiResponse.java  
      SalaryStructureResponse.java  
      StaffBankAccountResponse.java  
      PayrollResponse.java  
      PayrollSummaryResponse.java  
      SalaryCreditResponse.java  
      SalarySlipResponse.java  
      PayrollReportResponse.java  
      UserResponse.java  
      AttendanceSummaryResponse.java  
    mapper/  
      SalaryStructureMapper.java  
      StaffBankAccountMapper.java  
      PayrollMapper.java  
      PayrollPaymentBatchMapper.java  
      SalarySlipMapper.java  
    enums/  
      PayrollStatus.java  
      SalaryStructureStatus.java  
      SalaryComponentType.java  
      PaymentMode.java  
      SalarySlipStatus.java  
      PayrollBatchStatus.java  
      BankAccountStatus.java
```

```

exception/
    GlobalExceptionHandler.java
    ResourceNotFoundException.java
    InvalidStatusException.java
    BusinessException.java

    AccessDeniedException.java
    DuplicateResourceException.java
security/
    PayrollSecurityConfig.java
    JwtAuthenticationFilter.java
    CurrentUserProvider.java
    BranchAccessValidator.java
    RoleBasedAccessService.java
    MethodSecurityConfig.java
    AuthenticationEntryPointHandler.java
    AccessDeniedHandlerImpl.java
config/
    OpenApiConfig.java
    JpaAuditingConfig.java
    FeignClientConfig.java
    KafkaProducerConfig.java
    KafkaConsumerConfig.java
    KafkaTopicConfig.java
    AppConfig.java
client/
    UserServiceClient.java
    AttendanceServiceClient.java
    NotificationServiceClient.java
event/
    PayrollGeneratedEvent.java
    SalaryCreditedEvent.java
    SalaryPaymentFailedEvent.java
    SalarySlipGeneratedEvent.java
producer/
    PayrollEventProducer.java
consumer/
    AttendanceMarkedEventConsumer.java
constants/
    PayrollConstants.java
    KafkaTopicConstants.java
    HeaderConstants.java
util/
    PayrollUtil.java
    DateRangeUtil.java
validation/
    PayrollValidationService.java
audit/
    AuditLogRequest.java
    PayrollAuditHelper.java
src/main/resources/
    application.yml
    application-dev.yml
    application-prod.yml
    logback-spring.xml
db/migration/
    V1__create_core_tables.sql
    V2__create_indexes.sql
    V3__insert_default_data.sql
src/test/java/com/agacostays/payroll/
    controller/
    service/
    repository/
    mapper/
    security/
    producer/
    consumer/

```

13. Analytics Service

Field	Value
Application folder	analytics-service
Application class	AnalyticsServiceApplication.java
Base package	com.agacostays.analytics
Database	analytics_db
Purpose	Dashboards, projections, KPIs and exports

Workflow

Kafka events -> projection persistence -> role/branch-filtered query -> KPI/chart response -> PDF/Excel/CSV export.

Package Structure

```
analytics-service/  
  pom.xml  
  mvnw  
  mvnw.cmd  
  .mvn/wrapper/maven-wrapper.properties  
  Dockerfile  
  .dockerignore  
  README.md  
  src/main/java/com/agacostays/analytics/  
    controller/  
      HotelAnalyticsController.java  
      RestaurantAnalyticsController.java  
      BranchAnalyticsController.java  
      ManagerAnalyticsController.java  
      RootAdminAnalyticsController.java  
      ReportExportController.java  
    service/  
      HotelAnalyticsService.java  
      RestaurantAnalyticsService.java  
      BranchAnalyticsService.java  
      ManagerAnalyticsService.java  
      RootAdminAnalyticsService.java  
      ReportExportService.java  
      AnalyticsProjectionService.java  
    service/impl/  
      HotelAnalyticsServiceImpl.java  
      RestaurantAnalyticsServiceImpl.java  
      BranchAnalyticsServiceImpl.java  
      ManagerAnalyticsServiceImpl.java  
      RootAdminAnalyticsServiceImpl.java  
      ReportExportServiceImpl.java  
      AnalyticsProjectionServiceImpl.java  
    repository/  
      AnalyticsQueryRepository.java  
      ReportRepository.java  
      HotelRevenueProjectionRepository.java  
      RestaurantSalesProjectionRepository.java  
      AttendanceProjectionRepository.java  
      PayrollProjectionRepository.java  
    entity/  
      Report.java  
      HotelRevenueProjection.java  
      RestaurantSalesProjection.java  
      AttendanceProjection.java  
      PayrollProjection.java  
      FeedbackProjection.java  
    dto/request/  
      AnalyticsFilterRequest.java  
      DateRangeRequest.java  
      ExportReportRequest.java  
      BranchRevenueRequest.java  
      RootAnalyticsRequest.java  
    dto/response/  
      ApiResponse.java  
      HotelDashboardResponse.java  
      RestaurantDashboardResponse.java  
      BranchRevenueResponse.java  
      RootDashboardResponse.java  
      ManagerDashboardResponse.java  
      ExportReportResponse.java  
      KpiResponse.java  
      ChartDataResponse.java  
    mapper/  
      ReportMapper.java  
      HotelRevenueProjectionMapper.java  
      RestaurantSalesProjectionMapper.java  
      AttendanceProjectionMapper.java  
      PayrollProjectionMapper.java  
      FeedbackProjectionMapper.java  
    enums/  
      ReportType.java  
      ReportFormat.java  
      DateRangeType.java  
      MetricType.java  
      AggregationType.java  
      ExportStatus.java  
    exception/  
      GlobalExceptionHandler.java  
      ResourceNotFoundException.java
```

```

InvalidStateException.java
BusinessRuleException.java
AccessDeniedException.java

DuplicateResourceException.java
security/
  AnalyticsSecurityConfig.java
  JwtAuthenticationFilter.java
  CurrentUserProvider.java
  BranchAccessValidator.java
  RoleBasedAccessService.java
  MethodSecurityConfig.java
  AuthenticationEntryPointHandler.java
  AccessDeniedHandlerImpl.java
config/
  OpenApiConfig.java
  JpaAuditingConfig.java
  FeignClientConfig.java
  KafkaProducerConfig.java
  KafkaConsumerConfig.java
  KafkaTopicConfig.java
  AppConfig.java
client/
  BookingServiceClient.java
  RestaurantServiceClient.java
  PaymentServiceClient.java
  AttendanceServiceClient.java
  PayrollServiceClient.java
event/
  AnalyticsProjectionUpdatedEvent.java
  ReportExportedEvent.java
producer/
  AnalyticsEventProducer.java
consumer/
  BookingCreatedEventConsumer.java
  PaymentSuccessEventConsumer.java
  FoodDeliveredEventConsumer.java
  AttendanceMarkedEventConsumer.java
  PayrollGeneratedEventConsumer.java
  FeedbackSubmittedEventConsumer.java
constants/
  AnalyticsConstants.java
  KafkaTopicConstants.java
  HeaderConstants.java
util/
  AnalyticsUtil.java
  DateRangeUtil.java
validation/
  AnalyticsValidationService.java
audit/
  AuditLogRequest.java
  AnalyticsAuditHelper.java
src/main/resources/
  application.yml
  application-dev.yml
  application-prod.yml
  logback-spring.xml
db/migration/
  V1__create_core_tables.sql
  V2__create_indexes.sql
  V3__insert_default_data.sql
src/test/java/com/agacostays/analytics/
  controller/
  service/
  repository/
  mapper/
  security/
  producer/
  consumer/

```

14. Support Feedback Service

Field	Value
Application folder	support-feedback-service
Application class	SupportFeedbackServiceApplication.java
Base package	com.agacostays.support
Database	support_db
Purpose	Support requests, assignments, hotel feedback and restaurant feedback

Workflow

Support/feedback request -> ownership/booking/order validation -> persistence -> staff assignment/status updates -> Kafka notification/analytics event.

Package Structure

```
support-feedback-service/
  pom.xml
  mvnw
  mvnw.cmd
  .mvn/wrapper/maven-wrapper.properties
  Dockerfile
  .dockerignore
  README.md
  src/main/java/com/agacostays/support/
    controller/
      CustomerSupportController.java
      FeedbackController.java
      RestaurantFeedbackController.java
      SupportAssignmentController.java
    service/
      CustomerSupportService.java
      FeedbackService.java
      RestaurantFeedbackService.java
      SupportAssignmentService.java
    service/impl/
      CustomerSupportServiceImpl.java
      FeedbackServiceImpl.java
      RestaurantFeedbackServiceImpl.java
      SupportAssignmentServiceImpl.java
    repository/
      CustomerSupportRequestRepository.java
      FeedbackRepository.java
      RestaurantFeedbackRepository.java
      SupportAssignmentRepository.java
    entity/
      CustomerSupportRequest.java
      Feedback.java
      RestaurantFeedback.java
      SupportAssignment.java
    dto/request/
      CreateSupportRequest.java
      UpdateSupportStatusRequest.java
      FeedbackRequest.java
      RestaurantFeedbackRequest.java
      AssignSupportRequest.java
      SupportSearchRequest.java
    dto/response/
      ApiResponse.java
      SupportRequestResponse.java
      FeedbackResponse.java
      RestaurantFeedbackResponse.java
      SupportAssignmentResponse.java
      BookingResponse.java
      RestaurantOrderResponse.java
      UserResponse.java
    mapper/
      CustomerSupportRequestMapper.java
      FeedbackMapper.java
      RestaurantFeedbackMapper.java
      SupportAssignmentMapper.java
    enums/
      SupportRequestType.java
      SupportPriority.java
      SupportStatus.java
      FeedbackType.java
      RatingType.java
      AssignmentStatus.java
    exception/
      GlobalExceptionHandler.java
      ResourceNotFoundException.java
      InvalidStatusException.java
      BusinessException.java
      AccessDeniedException.java
      DuplicateResourceException.java
    security/
      SupportFeedbackSecurityConfig.java
      JwtAuthenticationFilter.java
      CurrentUserProvider.java
      BranchAccessValidator.java
      RoleBasedAccessService.java
      MethodSecurityConfig.java
      AuthenticationEntryPointHandler.java
```

```

    AccessDeniedHandlerImpl.java
config/
    OpenApiConfig.java
    JpaAuditingConfig.java
    FeignClientConfig.java

    KafkaProducerConfig.java
    KafkaConsumerConfig.java
    KafkaTopicConfig.java
    AppConfig.java
client/
    BookingServiceClient.java
    RestaurantServiceClient.java
    NotificationServiceClient.java
    UserServiceClient.java
event/
    SupportRequestCreatedEvent.java
    SupportStatusChangedEvent.java
    FeedbackSubmittedEvent.java
    RestaurantFeedbackSubmittedEvent.java
producer/
    SupportFeedbackEventProducer.java
consumer/
    CheckoutCompletedEventConsumer.java
    FoodDeliveredEventConsumer.java
constants/
    SupportFeedbackConstants.java
    KafkaTopicConstants.java
    HeaderConstants.java
util/
    SupportFeedbackUtil.java
    DateRangeUtil.java
validation/
    SupportFeedbackValidationService.java
audit/
    AuditLogRequest.java
    SupportFeedbackAuditHelper.java
src/main/resources/
    application.yml
    application-dev.yml
    application-prod.yml
    logback-spring.xml
db/migration/
    V1__create_core_tables.sql
    V2__create_indexes.sql
    V3__insert_default_data.sql
src/test/java/com/agacostays/support/
    controller/
    service/
    repository/
    mapper/
    security/
    producer/
    consumer/

```

Kafka Event Strategy

Producer	Main events	Primary consumers
Auth	UserRegistered, PasswordResetRequested	User, Notification
User	ManagerCreated, StaffCreated, StaffAssigned, StaffTransferred	Notification, Analytics
Branch	BranchCreated, BranchUpdated, PhotoUpdated	Notification, Analytics
Room	RoomCreated, RoomPriceUpdated, RoomStatusChanged	Booking, Analytics
Booking	BookingCreated, Approved, Cancelled, CheckoutCompleted	Notification, Billing, Room, Analytics, Payment
Restaurant	FoodOrderPlaced, FoodReady, FoodDelivered	Notification, Billing, Analytics
Payment	PaymentSuccess, PaymentFailed, RefundProcessed	Booking, Billing, Restaurant, Notification, Analytics
Billing	FinalBillGenerated, InvoiceGenerated, BillPaid	Booking, Notification, Analytics
Attendance	AttendanceMarked, CheckedIn, CheckedOut	Payroll, Analytics
Payroll	PayrollGenerated, SalaryCredited	Notification, Analytics
Support	SupportRequestCreated, FeedbackSubmitted	Notification, Analytics

Final Generation Checklist

For each requested microservice ZIP, verify:

- [] Exact application folder, application class and base package
- [] Root Maven wrapper, pom.xml, Dockerfile, .dockerignore and README
- [] Every package and every file listed for that service
- [] Complete compilable Java code, no placeholders unless .gitkeep is explicitly listed
- [] DTO validation, entity mappings, repositories, mappers and enums
- [] Service interfaces, implementations, thin controllers and SLF4J logging
- [] Exceptions, ApiResponse, security classes and correct access validators
- [] Required Feign/WebClient clients only
- [] Required Kafka events, producer and consumers only
- [] application profile files, logging, Flyway migrations and tests
- [] No transition to another microservice until explicitly requested

Revision Addendum - APIs, Access Control and File Completeness

This revision adds service-by-service API endpoints, authorized actors, API purpose, and a completeness audit that supplements the earlier package structures. All supplemental files listed here are mandatory additions to the corresponding microservice structure.

Updated on: 2026-09-28

Actor and Access Model

Supported actors: ROOT_ADMIN, MANAGER, CUSTOMER, RECEPTIONIST, RESTAURANT_ADMIN, CHEF, SERVING_STAFF, HOUSEKEEPING_STAFF, SYSTEM. Public endpoints require no login. Owner endpoints require the authenticated customer to own the booking, payment, bill, order, support request or feedback. Branch-scoped staff endpoints require the requested branchId to match an active StaffBranchMapping. SYSTEM endpoints are callable only by trusted microservices or signed providers.

Access Enforcement Order

```
API Gateway authentication
-> role route validation
-> downstream JWT/claims validation
-> permission validation
-> branch access validation where branch-scoped
-> owner access validation where customer-owned
-> business-state validation
-> service operation
```

1. API Gateway Service APIs and Access

Method	Endpoint	Access	Purpose
GET	/actuator/health	PUBLIC / PLATFORM	Gateway and dependency health
GET	/api/gateway/health	PUBLIC / PLATFORM	Gateway status
ALL	/api/auth/**	PUBLIC	Route to Auth Service
GET	/api/cities/**	PUBLIC	Route public city APIs
GET	/api/hotel-branches/**	PUBLIC / AUTHENTICATED	Route branch and public catalog APIs
ALL	/api/root-admin/**	ROOT_ADMIN	Protected root administration routes
ALL	/api/manager/**	MANAGER / ROOT_ADMIN	Protected manager routes
ALL	/api/staff/**	AUTHORIZED STAFF	Protected staff routes
ALL	/api/bookings/**	CUSTOMER / AUTHORIZED STAFF	Route booking APIs
ALL	/api/payments/**	CUSTOMER / AUTHORIZED STAFF	Route payment APIs
ALL	/api/billing/**	CUSTOMER / AUTHORIZED STAFF	Route billing APIs
ALL	/api/restaurant/**	CUSTOMER / RESTAURANT STAFF	Route restaurant APIs
ALL	/api/attendance/**	STAFF / MANAGER	Route attendance APIs
ALL	/api/payroll/**	MANAGER / ROOT_ADMIN / STAFF SELF	Route payroll APIs
ALL	/api/analytics/**	MANAGER / ROOT_ADMIN	Route analytics APIs
ALL	/api/support/**	CUSTOMER / AUTHORIZED STAFF	Route support APIs

Mandatory Supplemental Files

```
enums/GatewayRouteType.java
enums/GatewayErrorCode.java
enums/RateLimitType.java
enums/DownstreamServiceName.java
config/RedisRateLimiterConfig.java
dto/response/ApiResponse.java
test/integration/GatewayRoutingIntegrationTest.java
```

2. Auth Service APIs and Access

Method	Endpoint	Access	Purpose
POST	/api/auth/register	PUBLIC	Register customer
POST	/api/auth/login	PUBLIC	Authenticate and issue tokens
POST	/api/auth/refresh-token	PUBLIC WITH REFRESH TOKEN	Issue new access token
POST	/api/auth/logout	AUTHENTICATED	Revoke refresh token
POST	/api/auth/validate-token	GATEWAY / SYSTEM	Validate JWT and return claims
POST	/api/auth/forgot-password	PUBLIC	Start password reset
POST	/api/auth/reset-password	PUBLIC WITH RESET TOKEN	Reset password
PUT	/api/auth/change-password	AUTHENTICATED	Change own password
POST	/api/auth/otp/send	PUBLIC / SYSTEM	Send OTP
POST	/api/auth/otp/verify	PUBLIC	Verify OTP
GET	/api/auth/me	AUTHENTICATED	Return current authenticated identity

Mandatory Supplemental Files

```
validation/AuthValidationService.java
audit/LoginAudit.java
scheduler/ExpiredTokenCleanupScheduler.java
config/RedisConfig.java
exception/AccountLockedException.java
exception/InvalidOtpException.java
dto/response/ApiResponse.java
```

3. User Service APIs and Access

Method	Endpoint	Access	Purpose
POST	/api/root-admin/managers	ROOT_ADMIN	Create manager
GET	/api/root-admin/managers	ROOT_ADMIN	List managers
GET	/api/root-admin/managers/{managerId}	ROOT_ADMIN	Get manager
PUT	/api/root-admin/managers/{managerId}	ROOT_ADMIN	Update manager
PUT	/api/root-admin/managers/{managerId}/status	ROOT_ADMIN	Change manager status
DELETE	/api/root-admin/managers/{managerId}	ROOT_ADMIN	Soft-delete manager
POST	/api/manager/staff	MANAGER / ROOT_ADMIN	Create staff
GET	/api/manager/staff	MANAGER / ROOT_ADMIN	List staff
GET	/api/manager/staff/{staffId}	MANAGER / ROOT_ADMIN	Get staff
PUT	/api/manager/staff/{staffId}	MANAGER / ROOT_ADMIN	Update staff
PUT	/api/manager/staff/{staffId}/assign-role	MANAGER / ROOT_ADMIN	Assign staff role
PUT	/api/manager/staff/{staffId}/status	MANAGER / ROOT_ADMIN	Update staff status
POST	/api/manager/hotel-branches/{branchId}/staff	MANAGER / ROOT_ADMIN	Assign staff to branch
PUT	/api/manager/staff/{staffId}/transfer-branch	MANAGER / ROOT_ADMIN	Transfer staff
GET	/api/manager/customers	MANAGER / ROOT_ADMIN	List customers
PUT	/api/manager/customers/{customerId}/status	MANAGER / ROOT_ADMIN	Update customer status
POST	/api/roles	ROOT_ADMIN	Create role
GET	/api/roles	ROOT_ADMIN / MANAGER	List roles
POST	/api/permissions	ROOT_ADMIN	Create permission
POST	/api/permissions/roles/{roleId}	ROOT_ADMIN	Assign permission to role
GET	/api/profile/me	AUTHENTICATED	Get own profile
PUT	/api/profile/me	AUTHENTICATED	Update own profile

Mandatory Supplemental Files

```

scheduler/InactiveMappingCleanupScheduler.java
exception/FeignClientException.java
config/FeignErrorDecoder.java
dto/response/PageResponse.java
audit/UserAuditLog.java
test/integration/UserServiceIntegrationTest.java

```

4. Branch Service APIs and Access

Method	Endpoint	Access	Purpose
POST	/api/root-admin/cities	ROOT_ADMIN	Create city
GET	/api/cities	PUBLIC	List active cities
GET	/api/cities/{cityId}	PUBLIC	Get city
PUT	/api/root-admin/cities/{cityId}	ROOT_ADMIN	Update city
DELETE	/api/root-admin/cities/{cityId}	ROOT_ADMIN	Deactivate city
POST	/api/root-admin/hotel-branches	ROOT_ADMIN	Create branch
GET	/api/hotel-branches	PUBLIC	List branches
GET	/api/hotel-branches/{branchId}	PUBLIC	Get branch details
GET	/api/hotel-branches/city/{cityId}	PUBLIC	List city branches
PUT	/api/root-admin/hotel-branches/{branchId}	ROOT_ADMIN	Update branch
PUT	/api/root-admin/hotel-branches/{branchId}/status	ROOT_ADMIN	Change branch status
POST	/api/root-admin/hotel-branches/{branchId}/photos	ROOT_ADMIN / MANAGER	Upload branch photo
GET	/api/hotel-branches/{branchId}/photos	PUBLIC	List branch photos
DELETE	/api/root-admin/hotel-branches/{branchId}/photos/{photoId}	ROOT_ADMIN / MANAGER	Delete photo
POST	/api/manager/hotel-branches/{branchId}/receptionist-contacts	MANAGER	Add receptionist contact
GET	/api/hotel-branches/{branchId}/contacts	PUBLIC / AUTHENTICATED	Get branch contacts
POST	/api/manager/hotel-branches/{branchId}/emergency-contacts	MANAGER	Add emergency contact

Mandatory Supplemental Files

```
storage/StorageService.java
storage/impl/LocalStorageServiceImpl.java
storage/impl/CloudStorageServiceImpl.java
exception/StorageException.java
dto/response/PageResponse.java
test/integration/BranchServiceIntegrationTest.java
```

5. Room Service APIs and Access

Method	Endpoint	Access	Purpose
POST	/api/manager/hotel-branches/{branchId}/rooms	MANAGER	Create room
GET	/api/hotel-branches/{branchId}/rooms	PUBLIC	List branch rooms
GET	/api/hotel-branches/{branchId}/rooms/{roomId}	PUBLIC	Get room
PUT	/api/manager/hotel-branches/{branchId}/rooms/{roomId}	MANAGER	Update room
DELETE	/api/manager/hotel-branches/{branchId}/rooms/{roomId}	MANAGER	Soft-delete room
PUT	/api/manager/hotel-branches/{branchId}/rooms/{roomId}/status	MANAGER / RECEPTIONIST / HOUSEKEEPING_STAFF	Update room status
PUT	/api/manager/hotel-branches/{branchId}/rooms/{roomId}/price	MANAGER	Update room price
GET	/api/manager/hotel-branches/{branchId}/rooms/{roomId}/price-history	MANAGER / ROOT_ADMIN	Get price history
GET	/api/hotel-branches/{branchId}/rooms/availability	PUBLIC	Search availability by date
POST	/api/manager/hotel-branches/{branchId}/rooms/{roomId}/photos	MANAGER	Upload room photo
GET	/api/hotel-branches/{branchId}/rooms/{roomId}/photos	PUBLIC	List room photos
DELETE	/api/manager/hotel-branches/{branchId}/rooms/{roomId}/photos/{photoId}	MANAGER	Delete room photo

Mandatory Supplemental Files

```
storage/StorageService.java
storage/impl/LocalStorageServiceImpl.java
storage/impl/CloudStorageServiceImpl.java
consumer/CheckoutCompletedEventConsumer.java
exception/StorageException.java
dto/response/PageResponse.java
test/integration/RoomServiceIntegrationTest.java
```

6. Booking Service APIs and Access

Method	Endpoint	Access	Purpose
POST	/api/hotel-branches/{branchId}/bookings	CUSTOMER / RECEPTIONIST	Create booking
GET	/api/bookings/{bookingId}	OWNER / AUTHORIZED STAFF	Get booking
GET	/api/bookings/my-bookings	CUSTOMER	List own bookings
GET	/api/hotel-branches/{branchId}/bookings	MANAGER / RECEPTIONIST	List branch bookings
PUT	/api/bookings/{bookingId}	OWNER / RECEPTIONIST	Update eligible booking
PUT	/api/bookings/{bookingId}/approve	MANAGER / RECEPTIONIST	Approve booking
PUT	/api/bookings/{bookingId}/reject	MANAGER / RECEPTIONIST	Reject booking
PUT	/api/bookings/{bookingId}/cancel	OWNER / MANAGER / RECEPTIONIST	Cancel booking
POST	/api/bookings/{bookingId}/check-in	RECEPTIONIST / MANAGER	Complete check-in
POST	/api/bookings/{bookingId}/check-out	RECEPTIONIST / MANAGER	Complete checkout
GET	/api/bookings/{bookingId}/history	OWNER / AUTHORIZED STAFF	Get booking history
GET	/api/hotel-branches/{branchId}/bookings/date-range	MANAGER / RECEPTIONIST	Search bookings by date

Mandatory Supplemental Files

```
idempotency/IdempotencyService.java
idempotency/IdempotencyRecord.java
repository/IdempotencyRecordRepository.java
scheduler/BookingExpiryScheduler.java
exception/IdempotencyConflictException.java
dto/response/PageResponse.java
test/integration/BookingServiceIntegrationTest.java
```

7. Restaurant Service APIs and Access

Method	Endpoint	Access	Purpose
GET	/api/hotel-branches/{branchId}/restaurant	PUBLIC	Get restaurant details
PUT	/api/restaurant-admin/hotel-branches/{branchId}/restaurant	RESTAURANT_ADMIN / MANAGER	Update restaurant
POST	/api/restaurant-admin/hotel-branches/{branchId}/menu	RESTAURANT_ADMIN	Create menu item
GET	/api/hotel-branches/{branchId}/restaurant/menu	PUBLIC	List menu
GET	/api/hotel-branches/{branchId}/restaurant/menu/{menuItemId}	PUBLIC	Get menu item
PUT	/api/restaurant-admin/menu/{menuItemId}	RESTAURANT_ADMIN	Update menu item
PUT	/api/restaurant-admin/menu/{menuItemId}/status	RESTAURANT_ADMIN	Update availability
POST	/api/restaurant-admin/menu/{menuItemId}/photos	RESTAURANT_ADMIN	Upload dish photo
POST	/api/restaurant/orders	CUSTOMER / RECEPTIONIST	Place food order
GET	/api/restaurant/orders/{orderId}	OWNER / RESTAURANT STAFF	Get order
GET	/api/restaurant/orders/my-orders	CUSTOMER	List own orders
PUT	/api/restaurant-admin/orders/{orderId}/accept	RESTAURANT_ADMIN	Accept order
PUT	/api/restaurant-admin/orders/{orderId}/assign-chef	RESTAURANT_ADMIN	Assign chef
PUT	/api/kitchen/orders/{orderId}/preparing	CHEF	Mark preparing
PUT	/api/kitchen/orders/{orderId}/ready	CHEF	Mark ready
PUT	/api/serving/orders/{orderId}/picked-up	SERVING_STAFF	Mark picked up
PUT	/api/serving/orders/{orderId}/delivered	SERVING_STAFF	Mark delivered

Mandatory Supplemental Files

```

storage/StorageService.java
storage/impl/LocalStorageServiceImpl.java
scheduler/OrderTimeoutScheduler.java
exception/StorageException.java
dto/response/PageResponse.java
test/integration/RestaurantServiceIntegrationTest.java

```

8. Payment Service APIs and Access

Method	Endpoint	Access	Purpose
POST	/api/payments/orders	CUSTOMER / AUTHORIZED STAFF	Create gateway payment order
POST	/api/payments/verify	CUSTOMER / SYSTEM	Verify payment signature
GET	/api/payments/{paymentId}	OWNER / AUTHORIZED STAFF	Get payment
GET	/api/payments/my-payments	CUSTOMER	List own payments
GET	/api/manager/hotel-branches/{branchId}/payments	MANAGER	List branch payments
POST	/api/payments/webhooks/razorpay	PAYMENT GATEWAY	Receive signed webhook
POST	/api/payments/{paymentId}/refunds	MANAGER / ROOT_ADMIN / SYSTEM	Request refund
GET	/api/payments/{paymentId}/refunds	OWNER / AUTHORIZED STAFF	List refunds
GET	/api/refunds/{refundId}	OWNER / AUTHORIZED STAFF	Get refund
POST	/api/restaurant/payments	CUSTOMER / AUTHORIZED STAFF	Pay restaurant order
POST	/api/billing/payments	CUSTOMER / RECEPTIONIST	Pay final bill

Mandatory Supplemental Files

```
idempotency/IdempotencyService.java
entity/PaymentIdempotencyRecord.java
repository/PaymentIdempotencyRepository.java
config/PaymentGatewayProperties.java
exception/IdempotencyConflictException.java
scheduler/PaymentReconciliationScheduler.java
dto/response/PageResponse.java
test/integration/PaymentServiceIntegrationTest.java
```

9. Billing Service APIs and Access

Method	Endpoint	Access	Purpose
POST	/api/billing/bookings/{bookingId}/generate	RECEPTIONIST / MANAGER / SYSTEM	Generate final bill
GET	/api/billing/bookings/{bookingId}	OWNER / AUTHORIZED STAFF	Get final bill
GET	/api/billing/{billId}	OWNER / AUTHORIZED STAFF	Get bill
POST	/api/billing/{billId}/discount	MANAGER	Apply approved discount
POST	/api/billing/{billId}/pay	CUSTOMER / RECEPTIONIST	Initiate final bill payment
GET	/api/billing/{billId}/charges	OWNER / AUTHORIZED STAFF	List charges
POST	/api/invoices/bills/{billId}	RECEPTIONIST / MANAGER / SYSTEM	Generate invoice
GET	/api/invoices/{invoiceId}	OWNER / AUTHORIZED STAFF	View invoice metadata
GET	/api/invoices/{invoiceId}/download	OWNER / AUTHORIZED STAFF	Download invoice
GET	/api/manager/hotel-branches/{branchId}/bills	MANAGER	List branch bills

Mandatory Supplemental Files

```
storage/InvoiceStorageService.java
storage/impl/InvoiceStorageServiceImpl.java
entity/TaxConfiguration.java
repository/TaxConfigurationRepository.java
dto/response/PageResponse.java
test/integration/BillingServiceIntegrationTest.java
```

10. Notification Service APIs and Access

Method	Endpoint	Access	Purpose
POST	/api/internal/notifications/email	SYSTEM	Send email
POST	/api/internal/notifications/sms	SYSTEM	Send SMS
GET	/api/notifications/my-notifications	AUTHENTICATED	List own notifications
PUT	/api/notifications/{notificationId}/read	OWNER	Mark read
POST	/api/internal/checkout-reminders	SYSTEM / BOOKING SERVICE	Schedule checkout reminder
DELETE	/api/internal/checkout-reminders/{reminderId}	SYSTEM	Cancel reminder
POST	/api/root-admin/templates/email	ROOT_ADMIN	Create email template
PUT	/api/root-admin/templates/email/{templateId}	ROOT_ADMIN	Update email template
POST	/api/root-admin/templates/sms	ROOT_ADMIN	Create SMS template
GET	/api/root-admin/notifications/logs	ROOT_ADMIN	Search notification logs

Mandatory Supplemental Files

```
scheduler/CheckoutReminderScheduler.java
retry/NotificationRetryService.java
provider/EmailProvider.java
provider/SmsProvider.java
provider/impl/SmtEmailProvider.java
provider/impl/ApiSmsProvider.java
dto/response/PageResponse.java
test/integration/NotificationServiceIntegrationTest.java
```

11. Attendance Service APIs and Access

Method	Endpoint	Access	Purpose
POST	/api/attendance/check-in	STAFF	Self check-in
POST	/api/attendance/check-out	STAFF	Self check-out
GET	/api/attendance/my-attendance	STAFF	View own attendance
POST	/api/manager/staff/{staffId}/attendance	MANAGER	Manually mark attendance
PUT	/api/manager/attendance/{attendanceId}	MANAGER	Correct attendance
GET	/api/manager/hotel-branches/{branchId}/attendance	MANAGER	Branch attendance
GET	/api/manager/hotel-branches/{branchId}/working-staff	MANAGER / RECEPTIONIST	Currently working staff
GET	/api/manager/hotel-branches/{branchId}/absent-staff	MANAGER	Absent staff
GET	/api/attendance/staff/{staffId}/monthly	STAFF SELF / MANAGER	Monthly attendance

Mandatory Supplemental Files

```
scheduler/MissedCheckoutScheduler.java  
dto/response/PageResponse.java  
exception/InvalidCheckInLocationException.java  
test/integration/AttendanceServiceIntegrationTest.java
```

12. Payroll Service APIs and Access

Method	Endpoint	Access	Purpose
POST	/api/manager/salary-structures	MANAGER / ROOT_ADMIN	Create salary structure
GET	/api/manager/salary-structures/{staffId}	MANAGER / ROOT_ADMIN	Get salary structure
PUT	/api/manager/salary-structures/{staffId}	MANAGER / ROOT_ADMIN	Update salary structure
POST	/api/staff/bank-account	STAFF	Save own bank account
GET	/api/staff/bank-account	STAFF	Get masked own bank account
POST	/api/manager/payroll/generate	MANAGER / ROOT_ADMIN	Generate payroll batch
GET	/api/manager/payroll	MANAGER / ROOT_ADMIN	Search payroll
GET	/api/payroll/my-payroll	STAFF	View own payroll
POST	/api/manager/payroll/{payrollId}/credit	MANAGER / ROOT_ADMIN	Credit salary
GET	/api/payroll/{payrollId}/salary-slip	STAFF SELF / MANAGER	Download salary slip
GET	/api/manager/hotel-branches/{branchId}/payroll-report	MANAGER / ROOT_ADMIN	Branch payroll report

Mandatory Supplemental Files

storage/SalarySlipStorageService.java
storage/impl/SalarySlipStorageServiceImpl.java
scheduler/PayrollScheduleService.java
dto/response/PageResponse.java
exception/PayrollBatchNotFoundException.java
test/integration/PayrollServiceIntegrationTest.java

13. Analytics Service APIs and Access

Method	Endpoint	Access	Purpose
GET	/api/analytics/hotel-dashboard	MANAGER / ROOT_ADMIN	Hotel KPIs
GET	/api/analytics/restaurant-dashboard	RESTAURANT_ADMIN / MANAGER / ROOT_ADMIN	Restaurant KPIs
GET	/api/analytics/hotel-branches/{branchId}	MANAGER / ROOT_ADMIN	Branch analytics
GET	/api/analytics/manager-dashboard	MANAGER	Manager dashboard
GET	/api/analytics/root-admin-dashboard	ROOT_ADMIN	Global dashboard
GET	/api/analytics/revenue	MANAGER / ROOT_ADMIN	Revenue metrics
GET	/api/analytics/occupancy	MANAGER / ROOT_ADMIN	Occupancy metrics
GET	/api/analytics/attendance	MANAGER / ROOT_ADMIN	Attendance metrics
GET	/api/analytics/payroll	MANAGER / ROOT_ADMIN	Payroll metrics
POST	/api/analytics/reports/export	MANAGER / ROOT_ADMIN	Export report
GET	/api/analytics/reports/{reportId}	REQUESTING USER / ROOT_ADMIN	Get export status/download

Mandatory Supplemental Files

```
scheduler/ProjectionRebuildScheduler.java
storage/ReportStorageService.java
storage/impl/ReportStorageServiceImpl.java
dto/response/PageResponse.java
exception/ProjectionUpdateException.java
test/integration/AnalyticsServiceIntegrationTest.java
```

14. Support Feedback Service APIs and Access

Method	Endpoint	Access	Purpose
POST	/api/support/requests	CUSTOMER	Create support request
GET	/api/support/requests/my-requests	CUSTOMER	List own support requests
GET	/api/support/requests/{requestId}	OWNER / AUTHORIZED STAFF	Get request
PUT	/api/support/requests/{requestId}/status	MANAGER / RECEPTIONIST / ASSIGNED STAFF	Update status
POST	/api/support/requests/{requestId}/assign	MANAGER / RECEPTIONIST	Assign request
POST	/api/feedback	CUSTOMER WITH COMPLETED STAY	Submit hotel feedback
GET	/api/feedback/my-feedback	CUSTOMER	List own feedback
GET	/api/manager/hotel-branches/{branchId}/feedback	MANAGER	Branch feedback
POST	/api/restaurant/feedback	CUSTOMER WITH DELIVERED ORDER	Submit restaurant feedback
GET	/api/restaurant-admin/hotel-branches/{branchId}/feedback	RESTAURANT_ADMIN / MANAGER	Restaurant feedback
PUT	/api/feedback/{feedbackId}	OWNER	Update eligible feedback
DELETE	/api/feedback/{feedbackId}	OWNER / ROOT_ADMIN	Remove feedback

Mandatory Supplemental Files

dto/response/PageResponse.java
scheduler/SupportEscalationScheduler.java
exception/SupportEscalationException.java
test/integration/SupportFeedbackServiceIntegrationTest.java

Cross-Service Internal APIs

Source	Target	Internal API / Contract	Purpose
Gateway	Auth	POST /api/auth/validate-token	Validate JWT and retrieve claims
User	Branch	GET /api/internal/hotel-branches/{branchId}/exists	Validate branch
Booking	Room	GET /api/internal/rooms/{roomId}/availability	Validate room/date availability
Booking	User	GET /api/internal/customers/{customerId}	Validate customer
Restaurant	Booking	GET /api/internal/bookings/{bookingId}/active	Validate active stay
Payment	Booking/Billing/Restaurant	GET internal payable-resource endpoints	Validate payable amount and owner
Billing	Booking	GET /api/internal/bookings/{bookingId}/charges	Fetch stay charge data
Billing	Restaurant	GET /api/internal/bookings/{bookingId}/restaurant-charges	Fetch delivered food charges
Payroll	Attendance	GET /api/internal/staff/{staffId}/attendance-summary	Calculate payroll
Notification	User/Branch	GET internal contact endpoints	Resolve recipient contact
Analytics	Domain services	Kafka projections preferred	Avoid cross-database joins

Standard Files Required in Every Business Microservice

```
pom.xml
mvnw
mvnw.cmd
.mvn/wrapper/maven-wrapper.properties
Dockerfile
.dockerignore
README.md
src/main/resources/application.yml
src/main/resources/application-dev.yml
src/main/resources/application-prod.yml
src/main/resources/logback-spring.xml
src/main/resources/db/migration/V1__create_core_tables.sql
src/main/resources/db/migration/V2__create_indexes.sql
src/main/resources/db/migration/V3__insert_default_data.sql
src/test/java/<base-package>/controller/
src/test/java/<base-package>/service/
src/test/java/<base-package>/repository/
src/test/java/<base-package>/mapper/
src/test/java/<base-package>/security/
src/test/java/<base-package>/producer/
src/test/java/<base-package>/consumer/
src/test/java/<base-package>/integration/
```

Final Completeness Rule

The original package structures plus the mandatory supplemental files in this addendum form the complete generation contract. A coding agent must generate every listed file for the selected microservice, must not replace service-specific files with generic placeholders, and must stop after producing that one microservice ZIP.